Mixes and graph covers: complete manuscript costs

Two originals, both entire common papers, and both complete applications

This comparison enlarges the existing Mixes topic. It retains all 21 pages of Cunningham and Hubard's mix-criteria article (F005) and all 12 pages of Leemans and Toledo's faithful-maniplex article (L016). Their corrected principal contracts have distinct scientific acceptance. The measurement counts two originals, not their 54 detailed source dispositions.

One whole-topic comparison. The primary coordinate is literal untrimmed PDF extraction, NFC characters without whitespace. It includes title matter, abstracts, references, page furniture and arXiv banners on both sides.

Complete manuscript	Pages	Characters
F005 source	21	57753
L016 source	12	35506
Common simultaneous-action theory U_0	5	12615
Common cover theory U_1	4	11641
F005 local application A_1	8	20788
F005 exceptional application A_2	4	10289
L016 application A_3	6	15125

$$R_{\text{manuscripts}} = \frac{57753 + 35506}{12615 + 11641 + 20788 + 10289 + 15125} = \frac{93259}{70458} = 1.323611.$$

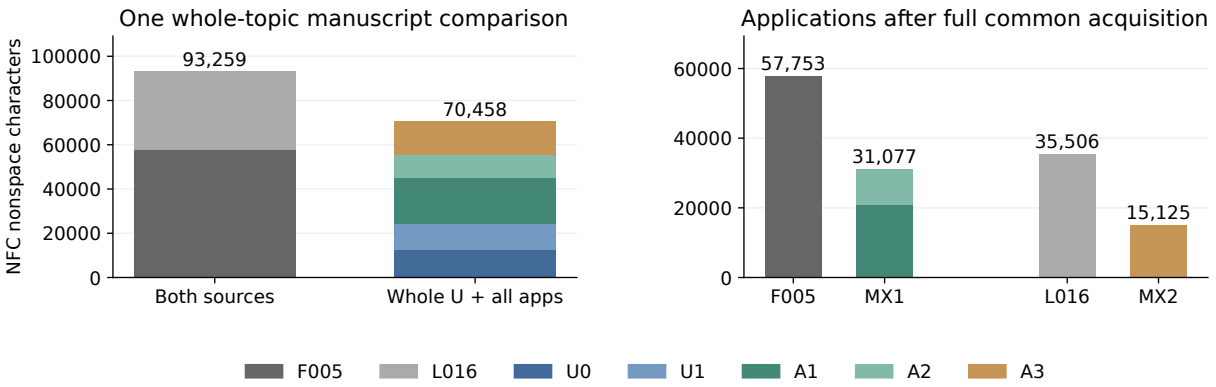
Both common papers are charged once in this topic. Both F005 papers remain one complete application, including its exceptional correction. No overlap is deducted, and no common premise is imported free from another topic.

Conditional comparisons. Only after acquiring *both entire* common papers, *both whole required finite certificates*, and all retained mathematical inputs, the individual source/application comparisons are

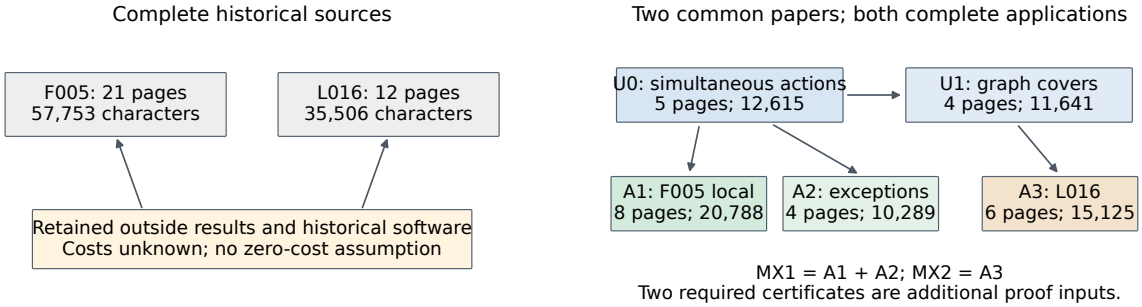
$$r_{\text{F005}} \mid (U, P) = \frac{57753}{20788 + 10289} = 1.858384, \qquad r_{\text{L016}} \mid (U, P) = \frac{35506}{15125} = 2.347504.$$

They do not waive the common acquisition in the whole-topic comparison. The written replacement is 22801 characters shorter in this primary representation. This measures manuscript length, not total research effort, minimum proof complexity or historical originality.

Symmetric software boundary. Optional validation is excluded on both sides. Neither fixed source contains an executable listing. Their mathematical tables, formulas, written algorithms and references to software remain counted. Unprovided historical software has unknown cost. Therefore no software-inclusive compression factor is calculated; the required programs remain essential separately disclosed proof inputs, rather than being reclassified as optional.



Both axes start at zero. The left panel stacks all source and replacement manuscripts. The right panel shows the two conditional comparisons; its application bars presuppose the whole common and required-certificate acquisition. Neither panel charges optional validation on only one side.



Arrows record paper-level proof reuse; box size is not a complexity measure. Full common papers are paid once.

The common argument is reused across simultaneous actions and equivariant graph covers. The paper-level arrows show that reuse, not an exhaustive historical theorem graph. Every displayed manuscript is counted in full; box dimensions and node counts are not complexity measures. Required finite certificates and retained external mathematical inputs remain explicit even though their full historical costs are unavailable.

Scope. F005 retains its five literal counterexample targets and two convention-qualified targets, with corrected rooted criteria; its three final problems remain unanswered. L016's literal Lemma 3.5 is disproved and repaired under faithfulness. Proposition 5.4 correctly says *without* the component intersection property. Its principal families and rank-three result are recovered; ancillary smallest/atlas counts and further-cover questions remain outside that contract. These qualifications do not prune either full source.

Whole programs and representation limits

The two required finite certificates are the complete `check_coprime.g` and `base_certificate.g`. Their raw primary cost is 2907 characters; the complete two-page printed alternative costs 3431. Printed and raw versions represent the same files and are never added together. The source-side counterpart remains unknown, so neither inventory produces an additional ratio.

Separate complete raw-file inventory	Files	Characters
Required finite certificates	2	2907
Optional validation	4	19467
Operational drivers	5	21999
Scoped Lean files	2	4181

All four optional files remain visible. The three-page new optional appendix costs 7285 characters; the two old mixed appendices cost 8813 and 7212 and remain historical inventories. They are not extra required charges or current optional-inclusive comparisons. The two Lean files provide six generic declarations, four new and two reused. They do not formalize all geometry, the finite presentations or either complete source.

Unit sensitivity. Complete untrimmed extraction under NFC gives:

Unit	Whole topic	F005 conditional	L016 conditional
Whitespace tokens	1.497	2.026	2.784
Alphanumeric runs	1.505	2.064	2.766
Nonspace characters	1.324	1.858	2.348
Collapsed-space characters	1.355	1.889	2.424
Collapsed-space UTF-8 bytes	1.279	1.775	2.309

The JSON and CSV retain all five units under NFC and NFKC in raw-untrimmed, raw-selected and calibrated-selected forms: 30 coordinates and 90 exact fractions. Selection removes only the separately bound running headers and isolated folios. Calibration changes declared visible glyph representations, including script letters, brace pieces and arrow-tee components; the original raw extraction always remains available. Literal line-break hyphens remain.

All 180 earlier F005 PDF cells and 70 raw-file cells are retained exactly. The new five-document input contributes 150 PDF cells and 60 raw-file cells. The combined eleven-document inventory includes historical code PDFs and therefore is *not* a list of eleven denominator terms. The current manuscript denominator contains exactly the two common papers and three application papers shown on page 1.

Diagram labels are retained when extractable. Full graphic/layout cost, retained outside proofs, original software, runtime, engineering and complete formalization remain unknown rather than zero. Distinct input acceptance is bound; this numerical author packet still requires a separate arithmetic, graph and reader review. It creates no new source, question or novelty credit.